2D Ideal MHD Solver on the Orszag-Tang Vortex

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May 1, 2026

Outline

1. Introduction and Problem Motivation
2. Mathematical Model
3. Numerical Model
4. Validation and Results
5. Discussion of Results
6. Conclusions

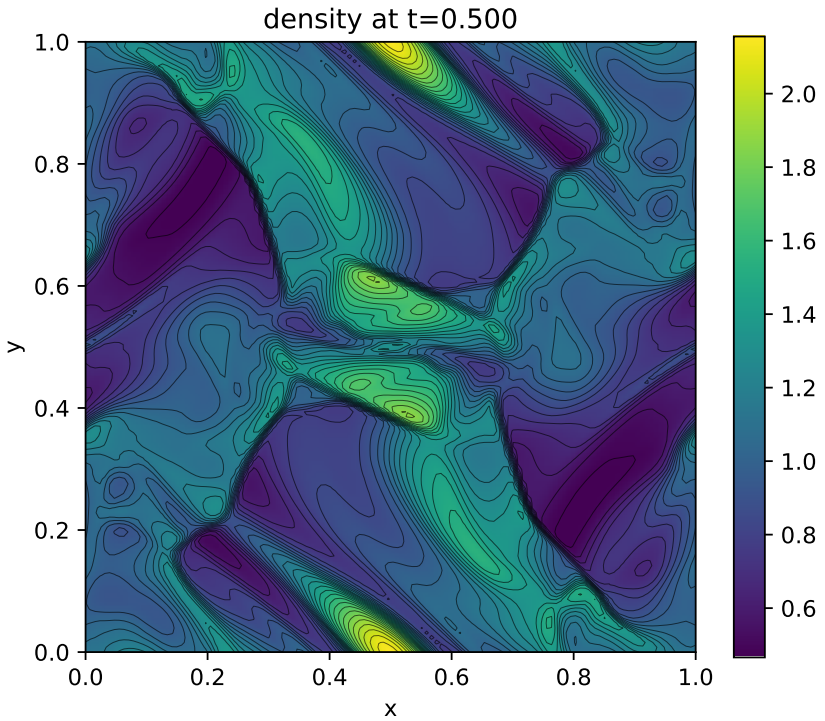
Why MHD

- Conducting fluid in motion = distributed current; *Lorentz force* $\mathbf{J} \times \mathbf{B}$ acts on every cell [1, 2].
- *Frozen-in flux*: in the ideal limit, field lines move with the fluid.
- Applies wherever plasma is collisional and magnetic Reynolds is large: solar wind, tokamaks, astrophysical jets, magnetospheres.
- New physics vs Euler:
  - *Alfven* waves—transverse, like waves on a tense string.
  - *Fast/slow magnetosonic*—sound waves modified by magnetic pressure.
  - Maxwell stresses $-B_i B_j$ in the momentum flux.
  - No-monopoles constraint $\nabla \cdot B = 0$.

Project goal: 2D ideal MHD finite-volume solver in Python, validated on Orszag-Tang.

Why Orszag-Tang

- Canonical 2D MHD benchmark.
- All-sinusoidal ICs on $[0, 1]^2$, periodic on every side.
- No geometry, no density jump, no outflow BCs.
- If the result is wrong, it is the solver, not the setup.
- Reference: [3, 4].



Ideal MHD in conservative form

$$\frac{\partial U}{\partial t} + \frac{\partial F}{\partial x} + \frac{\partial G}{\partial y} = 0, \quad U = (\rho,\ \rho\mathbf{v},\ \mathbf{B},\ E)^T$$

with $E = p/(\gamma - 1) + \frac{1}{2}\rho|\mathbf{v}|^2 + \frac{1}{2}|\mathbf{B}|^2$, $p_T = p + \frac{1}{2}|\mathbf{B}|^2$ [5].

- 8 conserved variables; all 3 velocity + field components, even in 2D.
- New vs Euler: magnetic pressure $\frac{1}{2}|B|^2$ in p_T, magnetic tension $-B_iB_j$, magnetic energy in E.
- Induction-equation flux $uB_y - vB_x$ is the motional EMF $-\mathbf{v} \times \mathbf{B}$: drags field lines with the flow.
- Seven-wave fan: 2 fast, 2 Alfven, 2 slow, 1 entropy [6, 7].
- Ideal-gas closure, $\gamma = 5/3$.

The divergence constraint

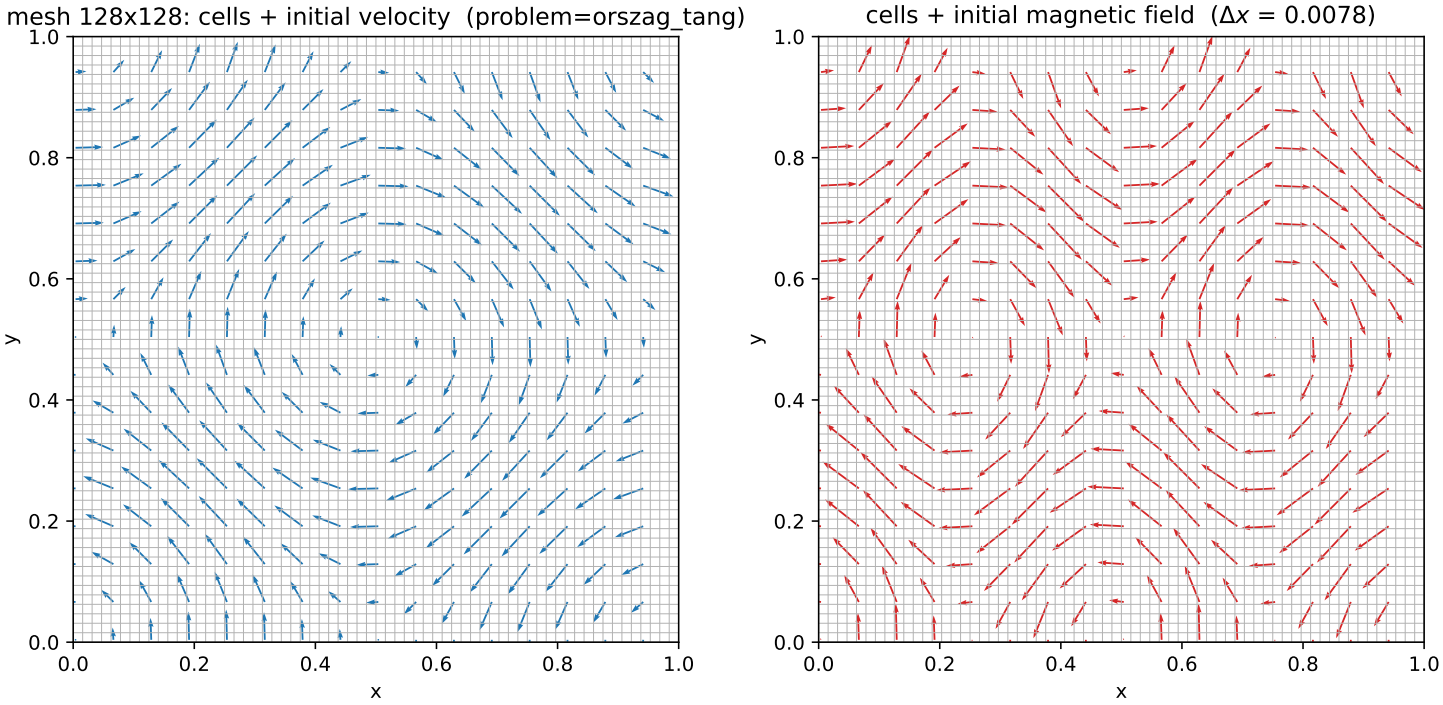
- $\nabla \cdot B = 0$ is Gauss's law for magnetism: no monopoles, no field-line sources or sinks.
- Exact MHD preserves this; discrete Godunov updates do not—reconstructed face fields leave a small divergence at every cell center [8, 5].
- Why it matters: the Lorentz force splits as

$$(\nabla \times B) \times B = (B \cdot \nabla)B - \tfrac{1}{2}\nabla|B|^2 - (\nabla \cdot B)B.$$

  The last term is zero analytically but becomes a spurious force *parallel to* B once $\nabla \cdot B$ drifts—momentum conservation breaks, run blows up.
- Remedies: constrained transport [8, 3], Powell source terms [5], or hyperbolic GLM cleaning [9].
- **This project uses GLM**: drops into an unstaggered FVM code, adds one scalar ψ.

Computational mesh

- Uniform 128×128 Cartesian grid, $\Delta x = \Delta y = 1/128$.
- Two ghost layers on every side (periodic fills for Orszag-Tang).
- Left: initial velocity; right: initial magnetic field.



Scheme at a glance

- Cell-centered Cartesian grid, two ghost layers [10, 11].
- MUSCL on primitives with minmod (MC, van Leer optional) [12].
- HLLD Riemann solver [7]; reduces to HLLC for $B = 0$.
- GLM ψ/B_x coupling baked into the Riemann solver; parabolic damping per RK stage [9].
- SSP-RK2 time integration [13], CFL $= 0.3$.

HLLD wave pattern

- Five-wave approximation to the full seven-wave MHD fan [7].
- Bounding speeds S_L, S_R from Davis-Einfeldt; entropy speed S_M from jump conditions.
- Two starred + two double-starred states across the rotational Alfven discontinuities.
- Positivity-preserving for physical inputs.

<table>
<tr><td>L</td><td>L^*</td><td>L^{**}</td><td>R^{**}</td><td>R^*</td><td>R</td></tr>
<tr><td colspan="2">S_L</td><td colspan="2">S_L^*</td><td>S_M</td><td>S_R</td></tr>
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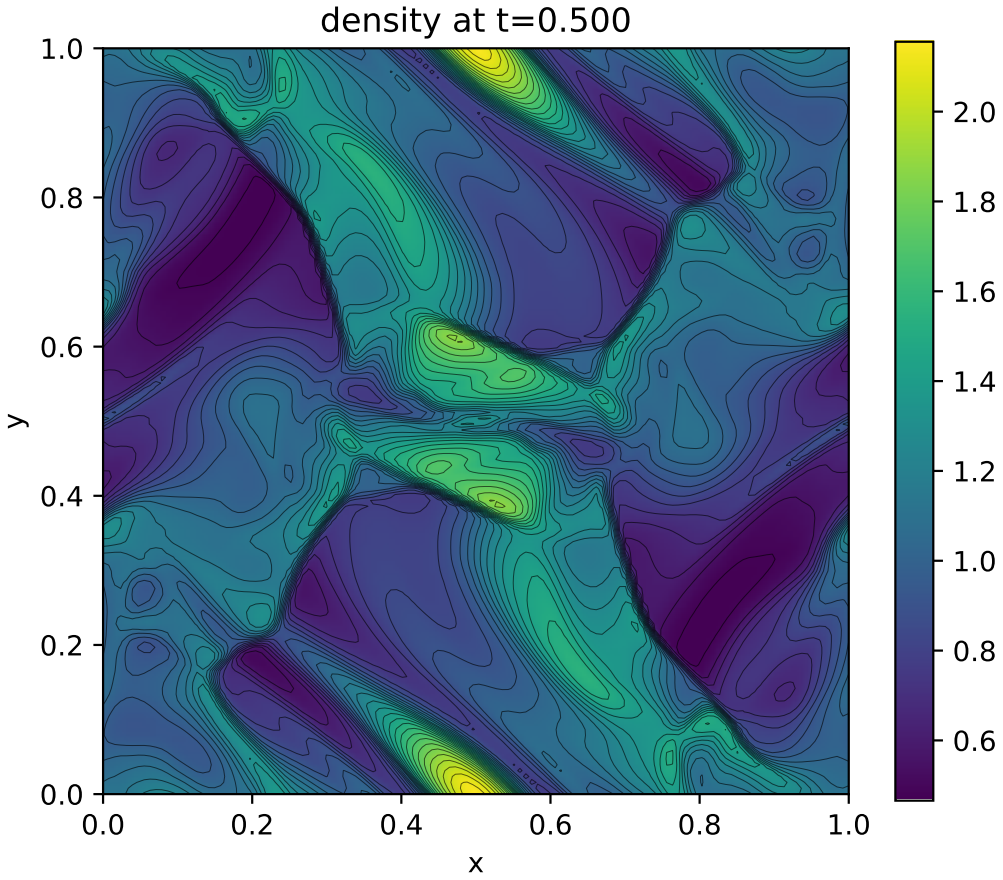
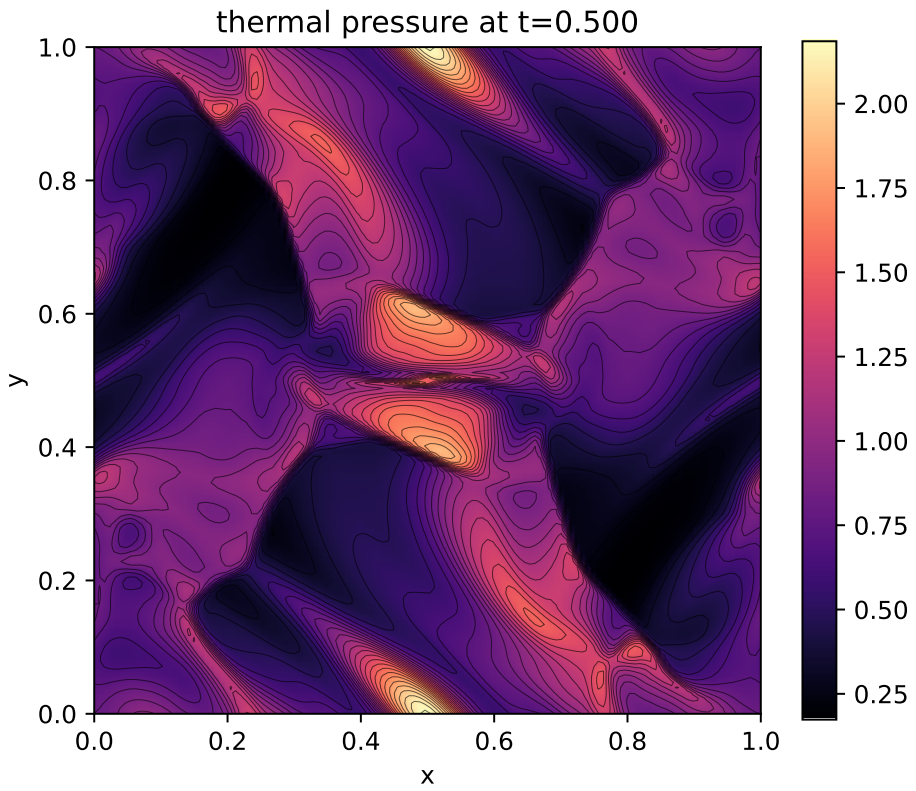
Wave regions across the x-face.

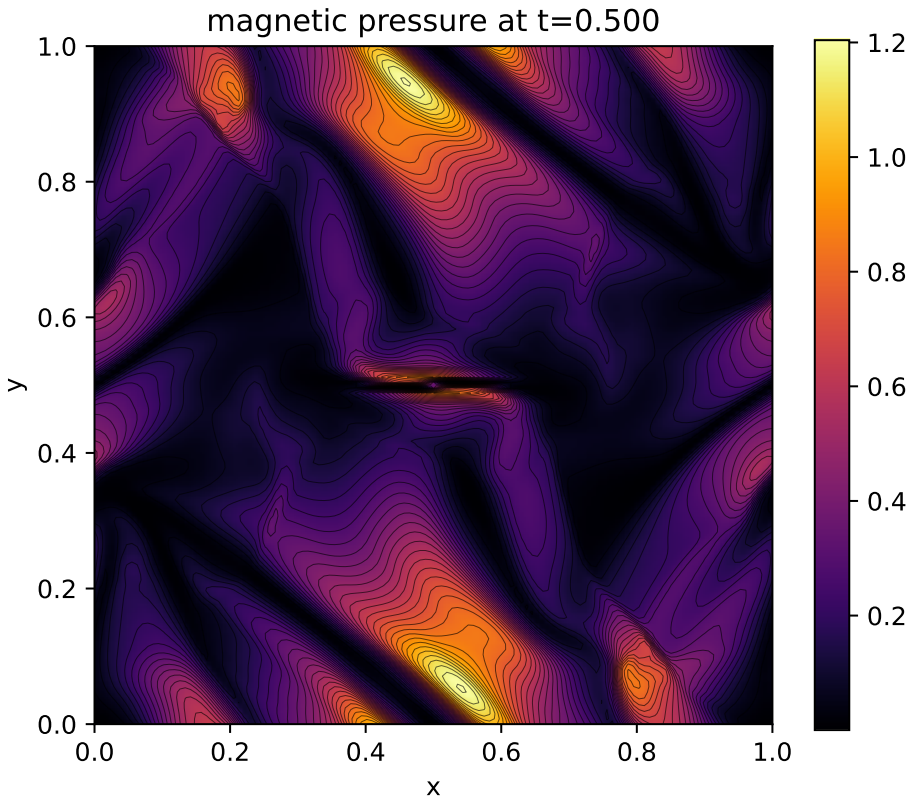
Density at t = 0.5, 128^2



- Central bow-tie current sheet.
- Four symmetric oblique shocks bounding low-density bays.
- Qualitative match with [3, 4] at the same time slice.

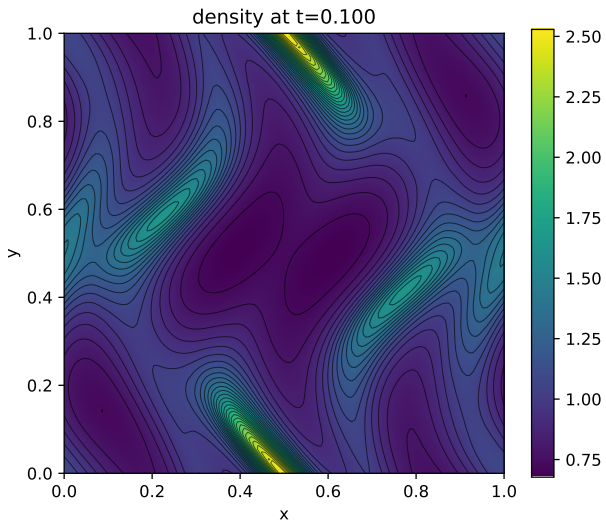
Thermal and magnetic pressure, t = 0.5



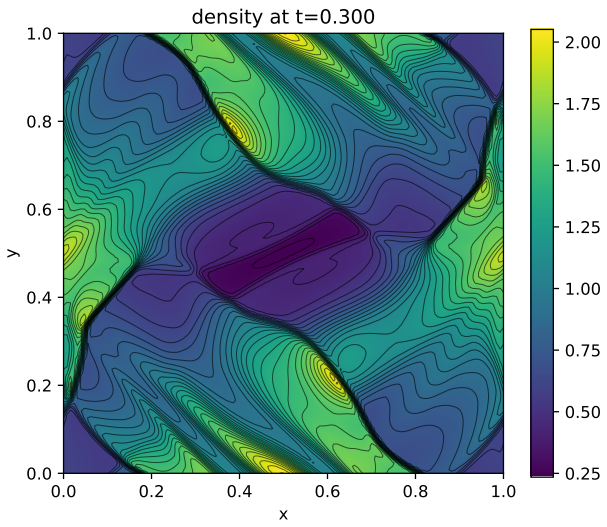


- Thermal pressure tracks density with extra shock heating near the current sheet.
- Magnetic pressure peaks at flux pile-up in the four current-sheet arms.

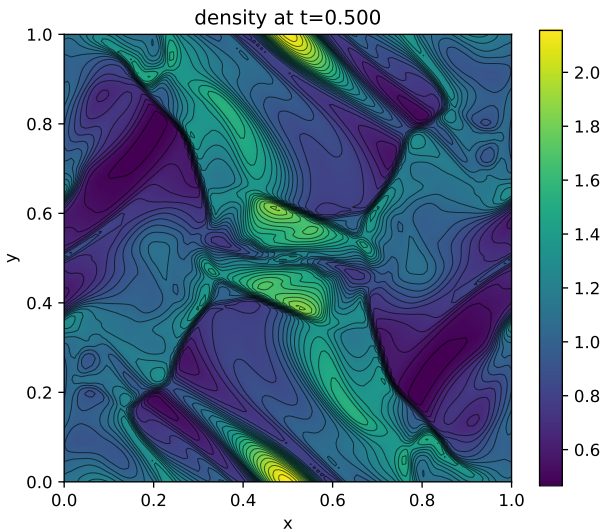
Formation of the current sheet



t = 0.1



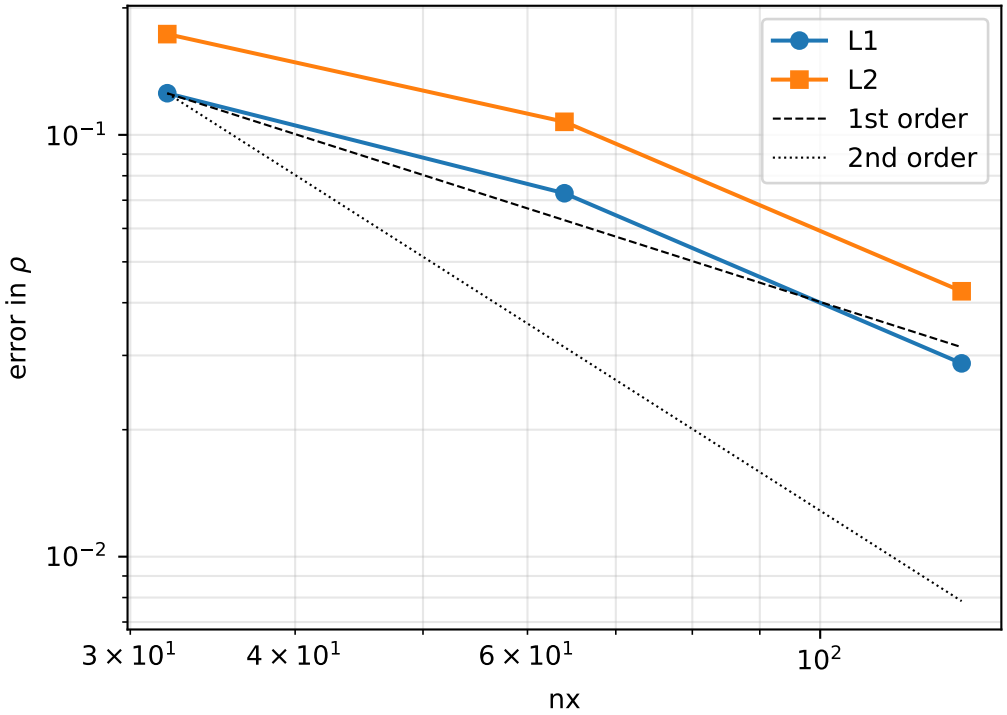
t = 0.3



t = 0.5

- t = 0.1: still smooth, the vortices are just forming.
- t = 0.3: diagonal shocks have formed; bow-tie compression starts.
- t = 0.5: fully-developed current sheet and bounding shocks.

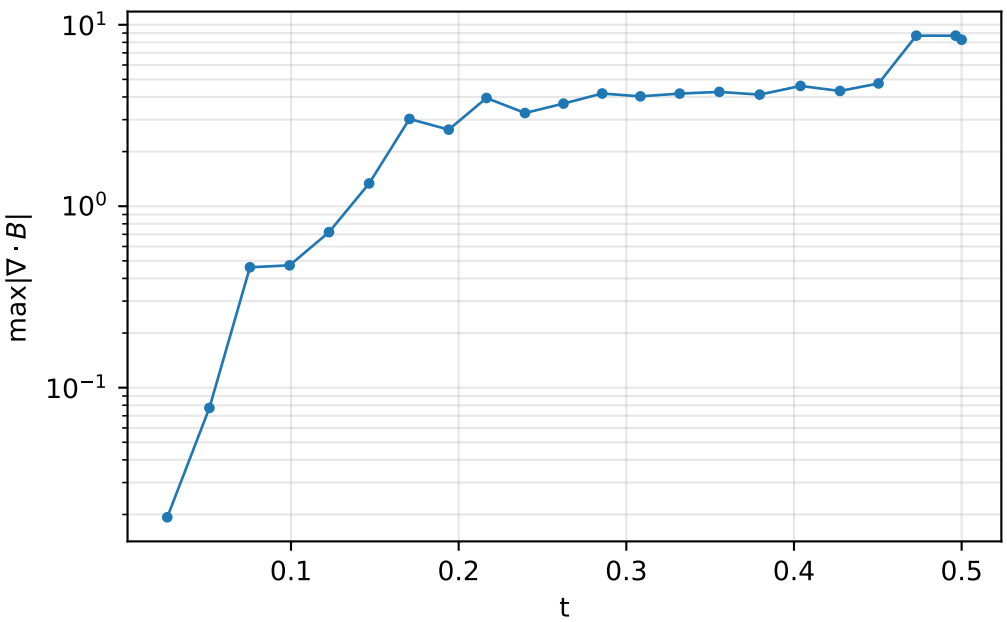
Convergence in density



nx	L^1	L^2
32	1.26e-1	1.73e-1
64	7.27e-2	1.07e-1
128	2.87e-2	4.26e-2

- $32 \to 64$: rate ≈ 0.79.
- $64 \to 128$: rate ≈ 1.33.
- Between 1st and 2nd order: expected for minmod on a shock-dominated flow [12].

Divergence control and conservation



- GLM [9] bounds $\max|\nabla \cdot B|$ once shocks form.
- Centered-difference $|\nabla \cdot B|$ is pessimistic; GLM-natural measure is smaller.
- Mass, momentum, energy conserved to machine precision ($\sim 10^{-16}$ per step, see `run.log`).

Accuracy vs limiter choice

- At 128^2 here: $\rho_{\text{min}} \approx 0.47$, $\rho_{\text{max}} \approx 2.16$.
- Athena++ with MC at the same resolution: $\rho_{\text{min}} \sim 0.1$, $\rho_{\text{max}} \sim 3.5$ [4].
- Gap is the limiter: minmod is the most diffusive TVD choice and smears current sheets [12].
- MC or van Leer would close it—but cost robustness in low-beta cells.

Why between first and second order

- MUSCL is second-order in smooth regions [12].
- TVD limiters drop to first order at extrema and at shocks.
- Orszag-Tang at $t = 0.5$ is shock-dominated $\Rightarrow$ global rate between 1 and 2.
- Observed: ≈ 0.8 coarse (shocks dominate), ≈ 1.3 fine (smooth regions start to dominate).

Limitations

- Vectorized NumPy: 128^2 in $\sim 7\,\mathrm{s}$; 256^2 is the practical ceiling on a laptop.
- No constrained-transport option—GLM is simpler but less strict on $\nabla \cdot B$ [8].
- 2D Cartesian; no AMR.
- Linear reconstruction (MUSCL); no WENO, no higher-order time integration.

Conclusions

- 2D ideal MHD solver from scratch: Godunov FVM, MUSCL-minmod, HLLD [7], GLM [9], SSP-RK2 [13].
- Orszag-Tang at 128^2 reproduces the canonical $t = 0.5$ density/pressure pattern [3, 4].
- Mass, momentum, energy conserved to machine precision; GLM bounds $\max|\nabla \cdot B|$.
- Convergence rate between 1 and 2, as expected for MUSCL-minmod on shock-dominated flow.

Natural extensions: constrained transport, resistive MHD, 3D, AMR, Numba/JAX.

References I

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